

NIELIT Virtual Academy

COURSE PROSPECTUS

Name of the Course: Certificate course in Web Development

Course Code : IT78

Mode of Conduction: Online Live

Last date of registration: 18th April 2025

Starting Date: 21st April 2025

Duration: 210 hours

Objective of the Course

This course equips participants with hands-on knowledge of HTML5, CSS, JavaScript, Bootstrap, Django, and Linux fundamentals. It covers server-side programming with PHP, Django, and database connectivity with MySQL for both Linux and Windows, preparing them for real-world web and software development with industry-relevant skills.

Prerequisites

- ✓ Candidate must have latest computer/laptop with preferably 4 GB RAM or higher and Graphics Card (2 GB)
- ✓ Internet connection with good speed (preferably 2Mbps or higher)

Outcome of the Course:

Participants will gain skills to create responsive websites using HTML5, CSS, JavaScript, and Bootstrap, understand Linux fundamentals, and implement server-side programming with PHP and Django. They will also learn database integration with MySQL, enabling the development of dynamic and robust web applications for Linux and Windows environments.

Course Fee: Rs: 6000/-

Eligibility

- Final-year Polytechnic Diploma in CS/IT after class 10th
- Or 2-Year ITI Certificate in IT/ITeS after class 10th
- Or 12th pass

Methodology:

- ✓ Teaching Mode: Online
- ✓ Instructor-led live sessions
- ✓ Online/ Blended lab sessions
- ✓ Content access through LMS
- ✓ Recorded Session Available

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- Course Coordinator Name: Syed Sameer
Co-coordinator Name: Harish Ramachandran
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Course Structure:

Module-1: Introduction to the Internet to web Design

- What is the Internet?
- Understanding URLs and Domain Names
- Understanding HTTP and HTTPS
- Internet vs. World Wide Web
- What is Web Design?

Module-2: HTML5

- Introduction to HTML
- HTML Tags and Elements
- Working with Links
- Adding Images
- Lists
- Tables
- Forms and Input Elements
- Semantic HTML
- Linking CSS and JavaScript
- HTML Entities

Module-3: CSS

- Introduction to CSS
- CSS Syntax
- Selectors
- Colors and Backgrounds
- Text Styling
- Margins and Padding

- Borders
- The Box Model
- Display and Positioning
- Flexbox
- Responsive Design

Module-3: JavaScript

- Introduction to JavaScript
- JavaScript Syntax
- Variables and Data Type
- Operators
- Conditional Statements
- Loops
- Functions
- DOM Manipulation
- Events
- Arrays
- Objects
- Timers
- Fetch API (Basic)

Module-4: Bootstrap

- Introduction to Bootstrap
- Bootstrap Grid System
- Typography
- Colors and Backgrounds
- Buttons
- Forms
- Navbar
- Cards
- Tables
- Responsive Design

Module-5 Linux Introduction

- What is Linux?
- Linux Interface
- Getting Started with Linux
- Basic Linux Commands
- Linux File System
- Viewing and Editing Files
- File Permissions
- Checking System Status
- Basic Networking

Module-6 Server-side programming language PHP

- What is PHP?
- PHP Tags and Comments
- Variables and Data Types
- PHP Operators
- Conditionals
- Loops
- HTML Form with PHP
- Handling Form Data in PHP
- Functions, Arrays, and Strings
- File Handling and Database Basics
- Connecting to MySQL Database
- Performing Queries

Module-7: Database connectivity with PHP and MySQL (Linux/Window)

- Introduction to PHP, MySQL, and Setup
- Database Operations (CRUD)

Module-8: Introduction to Django

- What is Django?
- Features of Django
- Installing Django and setting up the environment
- Understanding Django's architecture: MTV (Model-Template-View)
- Creating your first Django project and app
- Understanding settings.py and managing configurations
- URL routing and patterns
- Views and HttpResponse
- Templates and template rendering
- Static files and media files
- Django's development server

Module-9: Models and Database

- Introduction to Django ORM (Object-Relational Mapping)
- Creating and working with models
- Migrations: Creating, applying, and managing
- Database queries (CRUD operations)
- Using the Django Admin interface

Module-11: Forms and User Input

- Creating and handling forms
- Form validation and error handling
- Model forms
- Working with GET and POST requests
- File upload handling

Module-12: Django Authentication

- Built-in authentication system overview
- User registration and login
- Logout functionality
- Password reset and change
- User roles and permissions
- Customizing the User model

Module-13: Django REST Framework (DRF) Basics

- Introduction to REST APIs
- Setting up Django REST Framework
- Serializers and Views in DRF
- CRUD operations with APIs
- Authentication and Permissions in DRF
- Working with third-party API integrations

Examination & Certification

Paper Name	Certified Web Developer					Total
Category	Theory		Internal Assignments	Project / Presentation / Assignment	Practical	
Marks	Theory Paper-1 Front End Web Development	100	30	30	90	350
	Theory Paper-2 Database & Backend	100				

Marks Range	Grade	Certificate Type
$\geq 85\%$	S	Graded
$\geq 75\% \& < 85\%$	A	Graded
$\geq 65\% \& < 75\%$	B	Graded
$\geq 55\% \& < 65\%$	C	Graded
$\geq 50\% \& < 55\%$	D	Graded
$< 50\%$	F	Participation
Attended the Course but not appeared in Examination	N	Participation